

Churn Risk and Retention Report

Validated customer churn scoring from public Telco churn data

Rows analyzed

7,043

Validated customer rows

Observed churn

26.5%

Historical target rate

High risk

539

Priority accounts

Top 20% capture

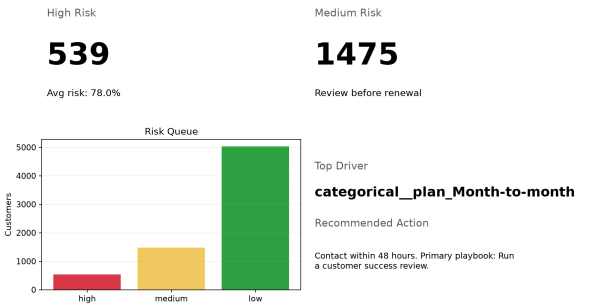
48.5%

Holdout churn capture

Executive Read

The model ranks customers by churn risk and turns that score into a retention queue. The highest-risk group should be reviewed first because the top 10% captures 29.2% of observed churners on the holdout split. This report separates model evidence from the full customer action file. Metrics, lift, calibration, and confidence intervals come from holdout customers; the scored queue covers the full customer base.

Churn Retention Executive Snapshot

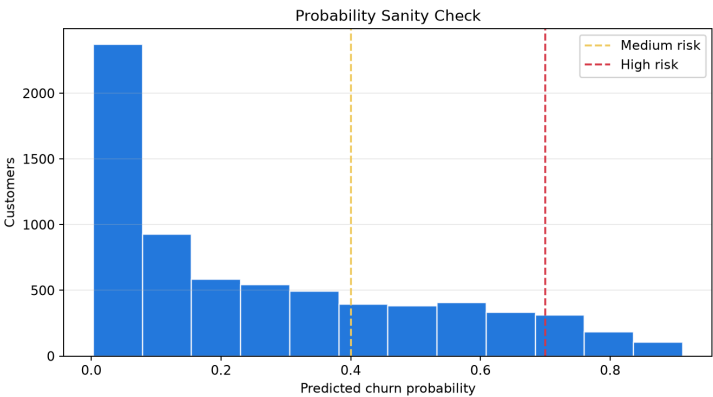


Dataset Summary

Metric	Value
rows	7043
columns	11
observed_churn_rate	26.54%
duplicate_customer_ids	0
leakage_warnings	0

Model Evidence

Metric	Value
roc_auc	0.8437
pr_auc	0.6564
precision	0.6358
recall	0.5134
f1	0.5680
brier_score	0.1360



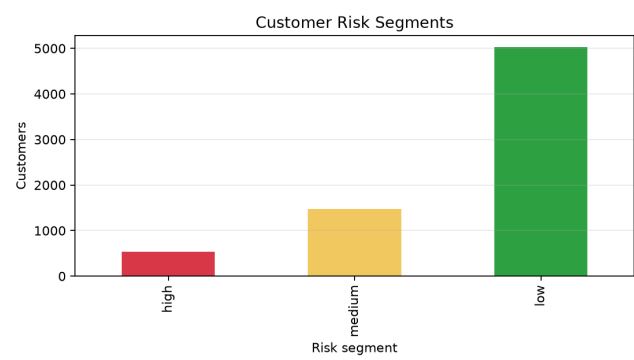
Lift and Calibration

decile	customers	churners	churn_rate	avg_prob	capture
1	212	164	0.7736	0.7440	0.2923
2	211	108	0.5118	0.5796	0.4848
3	211	99	0.4692	0.4534	0.6613
4	211	76	0.3602	0.3327	0.7968
5	212	45	0.2123	0.2302	0.8770

metric	estimate	ci_low	ci_high	samples
roc_auc	0.8437	0.8252	0.8620	300
pr_auc	0.6564	0.6105	0.6995	300

bin	customers	mean_pred	observed_rate	gap
1	212	0.7440	0.7736	0.0295
2	211	0.5796	0.5118	0.0677
3	211	0.4534	0.4692	0.0158
4	211	0.3327	0.3602	0.0274
5	211	0.2304	0.2133	0.0171

Risk Queue



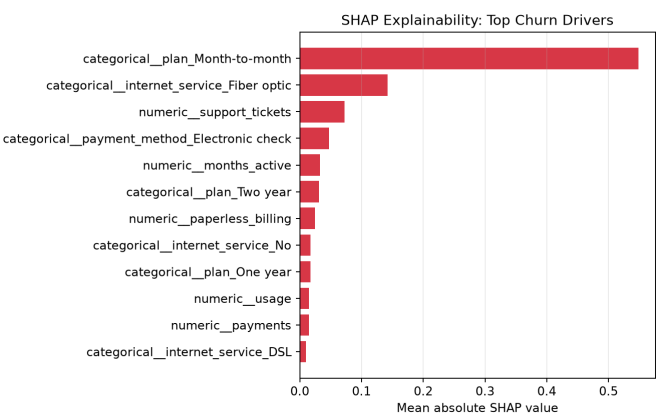
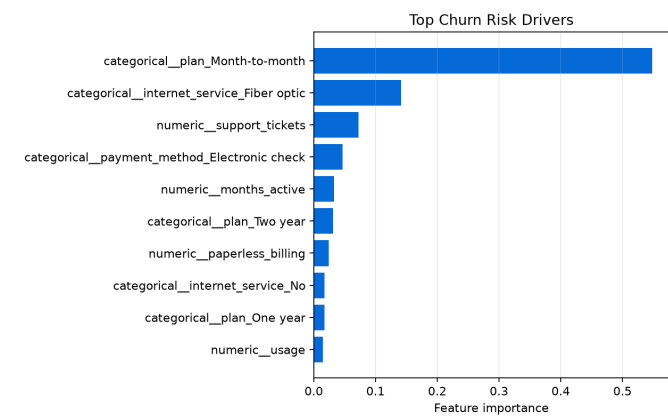
segment	customers	avg_prob
high	539	0.7801
medium	1475	0.5454
low	5029	0.1281

The PDF keeps actions short. Full recommended_action text is delivered in retention_recommendations.csv.

Top Recommended Accounts

customer_id	segment	risk	drivers	priority
7216-EWTRS	high	0.9119	general_engagement	1
0107-YHINA	high	0.9048	general_engagement	1
9497-QCMMS	high	0.8992	general_engagement	1
9300-AGZNL	high	0.8992	general_engagement	1
0295-PPHDO	high	0.8991	general_engagement	1

Driver Evidence



Scope

This is a one-time churn analysis package from a supplied CSV. It does not claim guaranteed accuracy, production CRM integration, scheduled retraining, or hosted monitoring. The useful output is the reproducible evidence bundle plus the customer-level retention queue.